

## Frontend Data Visualization: Dashboard Report

(Dynamic Views)

**Project Title:** *Supply Chain Resilience & Delivery Intelligence Dashboard : APL Logistics*

*(Proactive Attrition Logic & Predictive Modeling)*

**Industry Partner:** APL Logistics

**Organization:** Unified Mentor Pvt. Ltd. (Internship 2026)

**Tool Used:** Tableau Public Visuals & Tableau Formula

**Theme:** Professional Analytics Theme (As per Dashboard Visuals – Optimized for Clarity and Data-Driven Insights)

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### 1. Dashboard 1: Global Logistics Risk Map & KPI Overview (45% Threshold)

**Focus:** This dashboard highlights global hotspots of late deliveries & shows regions with the highest risk.

- **Key Logic & Visualization:**

- **Risk Thresholding:** Integrates a **Late Risk %** logic where **>45% is flagged as High Risk** and **<45% is considered within the Safe Threshold**.
- **Why 45% Threshold:** The 45% threshold was specifically selected over a 50% midpoint to foster a **proactive risk-aversion culture**. Setting the bar at 45% provides an "early warning buffer," allowing management to intervene when performance dips toward the 50% mark, rather than waiting for a failure to become statistically certain.
- **KPI Tracking:** Monitors 'Total Delayed Orders', 'Avg. Delivery Delay', and 'Early Orders Count'.
- **Purpose:** Provides an instant snapshot of global risk exposure, identifying whether the supply chain performance is operating above or below the 45% risk threshold.

### 2. Dashboard 2: Delivery Gap & Risk Trends Overview

- **Focus:** This dashboard analyzes delivery gap trends to identify risk progression and performance distribution patterns across order volumes.
- **Key Logic & Visualization:**
  - **Efficiency Analysis:** Maps 'Avg. Delivery Gap (Hours)' against shipping modes (Same Day, First Class, Second Class, Standard Class).
  - **Performance Segmentation:** Breakdowns by customer segments (Consumer, Corporate, Home Office).
- **Purpose:** Pinpoints exactly which shipping modes contribute most to the 'Late Risk %', allowing for targeted adjustments to logistics service levels.

### 2. Dashboard 3: 🌐 Geographical Scope & Regional Risk Distribution Insights

**Focus:** This dashboard breaks down risk levels by geographical scope and shows key KPIs for delivery performance.

- **Key Logic & Visualization:**
  - **Regional Heat Map:** Visualizes the intensity of delivery gaps across different sub-regions.
  - **Risk Breakdown:** Uses bubble and stacked bar charts to display the 'Geographical Scope Risk'.
- **Purpose:** Enables targeted intervention in high-risk regions that exceed the 45% risk threshold, ensuring focused resources are deployed where needed most.

#### 4. Dashboard 4: Predictive SLA Governance & Risk Mitigation

- **Focus:** SLA Governance: This dashboard includes end-to-end breach monitoring, regional criticality scoring, and segment-based performance insights to forecast and prevent delivery delays.
  - **Key Metrics & Formulas:**
    - **SLA Rate (%)** =  $1 - \text{Actual Breach Rate}$
    - **Actual Breach Rate (%)** =  $(\text{Total Delayed Orders} / \text{Total Orders}) \times 100$
  - **Risk Segmentation Logic:**
    - **Critical Risk:** >65% (Immediate intervention required).
    - **High Risk:** >45% (Monitoring and mitigation required).
    - **Safe Zone:** <45% (Stable operations).
  - **Purpose:** Forecasts performance risks before they result in contractual penalties, using the 65%/45% segmentation for clear operational prioritization.
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#### Executive Summary

The **APL Logistics Delivery Risk Prediction & Geographical Scope Dashboard** acts as the primary strategic instrument for maintaining supply chain reliability in an increasingly volatile global market. By integrating multi-dimensional data—ranging from shipping mode efficiency to regional delivery bottlenecks—this report provides a comprehensive view of logistics health.

The core of this dashboard is its **proactive risk framework**. By standardizing the 'Late Risk %' at a **45% threshold**, we move away from standard 50% benchmarks to create a "Safety Margin." This allows APL Logistics to identify performance decay before it reaches critical failure, effectively shifting the strategy from reactive "firefighting" to intelligent, data-driven prevention. The segmentation of SLA risks into Critical (>65%), High (>45%), and Safe (<45%) zones ensures that leadership can prioritize resource allocation to the most vulnerable regions, thereby minimizing financial penalties and bolstering customer trust. This dashboard does not just report on what happened; it provides the insights necessary to shape what will happen, ensuring operational excellence across all geographical scopes.

#### Future Scope

- **AI/ML Integration:** Incorporating advanced machine learning models (like Random Forest or XGBoost) directly into the dashboard to predict delivery delays with higher accuracy based on real-time weather and traffic data.

- **Automated Alerting System:** Implementing real-time email or SMS alerts for 'Critical Risk' (>65%) regions to notify warehouse managers instantly, reducing response time.
  - **Vendor Performance Tracking:** Expanding the scope to include specific transport vendor analytics, allowing for better negotiation and selection of reliable logistics partners.
  - **Cost-Impact Analysis:** Linking delivery delays directly to financial penalty costs to prioritize interventions based on ROI.
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## Conclusion

This comprehensive dashboard suite successfully transforms complex operational data into a clear, actionable hierarchy. The integration of defined risk thresholds and SLA formulas enables the management team to identify, prioritize, and resolve logistics bottlenecks with precision. By systematically addressing regions and shipping modes that fall outside the "Safe" threshold, APL Logistics can optimize its delivery performance, minimize the costs associated with SLA breaches, and significantly improve overall service reliability.